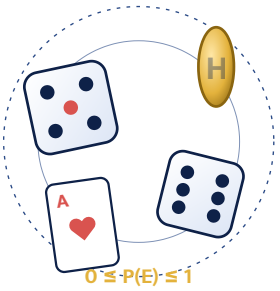
MT Classes, Motihari

Probability

Class 10 Maths, Chapter 14 (NCERT, CBSE 2026-27)



Coins, dice, cards, bags of balls and spinners: the whole chapter explained step by step beside every figure of the textbook, with a die you can roll, a 52-card deck explorer, the two-dice grid, a coin-toss lab, a spinner, a birthday surprise, a dart-drop experiment, real CBSE board questions 2023–2026, every NCERT question solved and a 20-question quiz.

📌 Chapter snapshot

- Chapter 14 of NCERT Class 10 Mathematics (reprint 2026-27): one section (14.1), 13 solved examples and **one exercise of 25 questions**.
- Part of the unit **Statistics and Probability** (11 marks in the 80-mark paper). Probability alone usually gives **4 to 7 marks**: two or three 1-mark MCQs, often an **Assertion–Reason**, and a 2- or 3-mark question on **two dice** or **cards**.
- Asked every year: **complement** (0.79 → 0.21, 2024), **cards** (red face card, 2025; lost card, 2024), **two dice** (difference 2, 2025; same number, 2026; sums, 2024), **bag of balls** (2023).
- Examples 10 and 11 and Exercise 20 (probability by length and area) are marked **"not from the examination point of view"**. Read them for understanding only.

14 Why a theoretical approach?

> *"The theory of probabilities and the theory of errors now constitute a formidable body of great mathematical interest and of great practical importance."*
>
> – R.S. Woodward

Suppose a coin is tossed **at random**. Before it lands we already know it can show only a **head** or a **tail**. Can we say how likely each one is **without** tossing it a thousand times? Yes, if we can make a sensible **assumption**. That is what this chapter is about.

🪙 Fair (unbiased) coin

Symmetrical, so there is no reason for it to fall more often on one side. We ignore it landing on its edge.

🎲 Fair die

For us, a die always means a fair die. Outcomes 1, 2, 3, 4, 5, 6 have the same chance of showing up.

🎯 Random

"Random toss" or "drawn at random" means the coin falls freely, or every item is equally likely to be picked, with no bias.

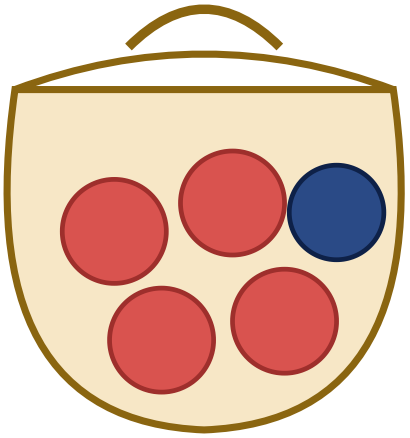
Equally likely outcomes

Outcomes are **equally likely** when each has the same chance of happening. Head and tail of a fair coin are equally likely; so are the six faces of a fair die.

Are outcomes always equally likely? No. A bag has **4 red balls and 1 blue ball**. You draw one ball without looking.

- The outcomes "red" and "blue" are **not** equally likely: red is more likely.
- But drawing **any one particular ball** is equally likely: each of the 5 balls has the same chance.
- So count the **balls**, not the **colours**: P(red) = $\frac{4}{5}$, P(blue) = $\frac{1}{5}$.

4 red + 1 blue = 5 balls



From here on, NCERT assumes that every experiment has equally likely outcomes.

Experimental probability vs theoretical probability

In Class IX you met the **experimental (empirical) probability**:

$$\mathbf{P(E) = \frac{\text{Number of trials in which E happened}}{\text{Total number of trials}}}$$

Needs the experiment to be repeated many times.

Repeating is not always possible: we cannot launch a satellite again and again to see how often it fails, or wait for many earthquakes to see how often a tall building falls. When we can assume the outcomes are **equally likely**, we can find the exact probability **without doing the experiment**. This is the **theoretical** (or **classical**) probability.



📜 A little history

Probability began in the **16th century** when the Italian physician and mathematician **J. Cardan** wrote the first book on it, *The Book on Games of Chance*. **James Bernoulli** (1654–1705), **A. de Moivre** (1667–1754) and **Pierre Simon Laplace** (1749–1827) made major contributions. Laplace gave the definition of theoretical probability in **1795**, and his *Théorie Analytique des Probabilités* (1812) is considered the greatest contribution by a single person to the subject. Today probability is used in biology, economics, genetics, physics, sociology, weather forecasts, insurance and cricket analytics.

14.1 Theoretical probability

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes of the experiment}}$$

where the outcomes of the experiment are equally likely. We simply call this the **probability** of E.

Experiment the action

Tossing a coin, throwing a die, drawing a card, picking a ball.

Outcome one possible result

Throwing a die: 1, 2, 3, 4, 5 or 6. All possible outcomes together are often called the **sample space** (S).

Event a collection of outcomes

"Getting an even number" = {2, 4, 6}. We name events with capital letters E, F, G.

Favourable outcomes outcomes that make E happen

For "even number", the favourable outcomes are 2, 4 and 6, so there are **3** of them.

✍️ The 4-line answer method (full marks every time)

1. Write the **total number of equally likely outcomes** (and list them if short).
2. Name the event: "Let E be the event '…' ".
3. List and count the **favourable outcomes**.
4. Write P(E) = favourable ÷ total, and **simplify** the fraction.

Worked examples from the book

Example 1 · one coin

Outcomes: H, T (2). E = getting a head: 1 favourable.

P(head) = $\frac{1}{2}$, P(tail) = $\frac{1}{2}$

Why is P(tail) also $\frac{1}{2}$? Tail is also 1 of the 2 equally likely outcomes.

Example 2 · three balls

A bag has 1 red, 1 blue, 1 yellow ball of the same size. Kritika takes one without looking. Total = 3.

P(Y) = P(R) = P(B) = $\frac{1}{3}$

Example 3 · one die

Total = 6. (i) E = number greater than 4 = {5, 6}. (ii) F = number ≤ 4 = {1, 2, 3, 4}.

P(E) = $\frac{2}{6}$ = $\frac{1}{3}$, P(F) = $\frac{4}{6}$ = $\frac{2}{3}$

14.1 Elementary and complementary events

Elementary events

An event having **only one outcome** of the experiment is an **elementary event**. In Example 1, "head" and "tail" are elementary; in Example 2, Y, R and B are elementary. In Example 3, E has 2 outcomes and F has 4, so they are **not** elementary.

⭐ Rule 1

The **sum of the probabilities of all the elementary events** of an experiment is **1**.

Coin: $\frac{1}{2} + \frac{1}{2} = 1$ · Bag: $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$ · Die: six times $\frac{1}{6} = 1$.

Complementary events: "not E"

In Example 3, "a number greater than 4" and "a number less than or equal to 4" together cover every outcome, with no overlap. F is the same as **"not E"**, written $\bar{E}$. We call $\bar{E}$ the **complement** of E, and E, $\bar{E}$ are **complementary events**.

$P(E) + P(\bar{E}) = 1 \Rightarrow P(\bar{E}) = 1 - P(E)$

Use it whenever "not", "at least one" or "different" is easier to count the other way round.



14.1 Impossible event, sure event and 0 ≤ P(E) ≤ 1

🚫 **Impossible event: P = 0**

Getting **8** on a die. No face shows 8, so there are 0 favourable outcomes: P = $\frac{0}{6}$ = **0**.

✅ **Sure (certain) event: P = 1**

Getting a number **less than 7** on a die. Every face qualifies: P = $\frac{6}{6}$ = **1**.

The number of favourable outcomes can never be negative and can never be more than the total, so

0 ≤ P(E) ≤ 1

A probability can never be negative and never more than 1. It may be written as a fraction, a decimal or a percentage (15% = 0.15 is fine).

The probability scale



Cards A deck of 52 playing cards

Card questions come in the board exam almost every year, so learn the deck by heart.

🃏 What is in the deck

- **52 cards** in **4 suits** of **13** cards each.
- **Black**: spades ♠ and clubs ♣ (26 cards). **Red**: hearts ♥ and diamonds ♦ (26 cards).
- Each suit: **ace, king, queen, jack, 10, 9, 8, 7, 6, 5, 4, 3, 2**.
- **Face cards**: kings, queens and jacks: 3 × 4 = **12** (6 red, 6 black). The ace is **not** a face card.

Type of card	Count	P
Any one suit (e.g. spade)	13	$\frac{1}{4}$
Red (or black)	26	$\frac{1}{2}$
Ace / king / queen / jack	4 each	$\frac{1}{13}$
Face cards	12	$\frac{3}{13}$
Red face cards	6	$\frac{3}{26}$
Red king	2	$\frac{1}{26}$
One named card (jack of hearts)	1	$\frac{1}{52}$
Number cards 2 to 10	36	$\frac{9}{13}$

Example 4 · one card from a well-shuffled deck

Well-shuffling ensures equally likely outcomes, so the total is 52 (one outcome for each card).

(i) E = the card is an **ace**: 4 aces. $\mathbf{P(E) = \frac{4}{52} = \frac{1}{13}}$

(ii) F = **not an ace**: 52 − 4 = 48 cards. $\mathbf{P(F) = \frac{48}{52} = \frac{12}{13}}$

Shortcut: F is the complement of E, so $P(F) = 1 - \frac{1}{13} = \frac{12}{13}$.

⚠️ Card traps

- "Face card" means **K, Q, J** only (12 cards). An ace is not a face card.
- "A red king" = king of hearts and king of diamonds = 2 cards. "The king of hearts" = 1 card.
- If some cards are **removed** or **lost**, the total changes: 52 − (cards removed). In CBSE 2024 a black card was lost, so the total became 51.

Ex. 5–9

Matches, birthdays, classrooms, bags and coins

Example 5 · tennis match

P(Sangeeta wins) = 0.62. Reshma winning is the complement (one of them must win).

P(Reshma) = 1 − 0.62 = 0.38

Example 6 · two friends' birthdays

Savita's birthday is any day; Hamida's can be any of 365 days (ignore leap years). Different birthdays: 365 − 1 = 364 days.

P(different) = $\frac{364}{365}$, P(same) = 1 − $\frac{364}{365}$ = $\frac{1}{365}$

Example 7 · class representative

40 students (25 girls, 15 boys); one name card is drawn.

P(girl) = $\frac{25}{40}$ = $\frac{5}{8}$, P(boy) = $\frac{15}{40}$ = $\frac{3}{8}$

Also P(boy) = 1 − P(girl) = 1 − $\frac{5}{8}$ = $\frac{3}{8}$.

Example 8 · marbles

3 blue, 2 white, 4 red: 3 + 2 + 4 = 9 marbles.

P(W) = $\frac{2}{9}$, P(B) = $\frac{3}{9}$ = $\frac{1}{3}$, P(R) = $\frac{4}{9}$

Check: $\frac{2}{9} + \frac{3}{9} + \frac{4}{9} = 1$.

Example 9 · two coins (₹1 and ₹2)

Outcomes: (H, H), (H, T), (T, H), (T, T): 4. At least one head: (H, H), (H, T), (T, H): 3.

P(at least one head) = $\frac{3}{4}$

Or: 1 − P(no head) = 1 − $\frac{1}{4}$ = $\frac{3}{4}$.

Example 12 · shirts

100 shirts: 88 good, 8 minor defects, 4 major defects. Jimmy accepts only good shirts; Sujatha rejects only major defects.

P(Jimmy) = $\frac{88}{100}$ = 0.88, P(Sujatha) = $\frac{96}{100}$ = 0.96

Sujatha accepts 88 + 8 = 96.

Tossing coins: draw a tree

Each coin doubles the number of outcomes: 1 coin → 2, 2 coins → 4, 3 coins → 8 (2^n for n coins). Tossing one coin 3 times is the same experiment as tossing 3 coins together.

Spinner (Fig. 14.5) and the fish tank (Fig. 14.4)



Fig. 14.4

Fig. 14.4 5 male + 8 female fish: P(male) = $\frac{5}{13}$ (Exercise 11)

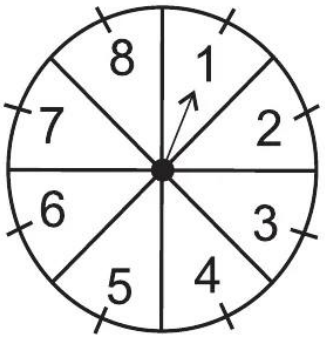


Fig. 14.5

Fig. 14.5 8 equally likely numbers (Exercise 12)

Birthdays: a surprise

Example 6 shows that two people share a birthday with probability only $\frac{1}{365}$. But in a bigger group, the chance that **some two** people share a birthday grows surprisingly fast.

Dice

Two dice: Example 13 and Fig. 14.3

Two dice, one **blue** and one **grey**, are thrown together. When the blue die shows 1, the grey die can show 1 to 6; the same for every number on the blue die. So there are **6 × 6 = 36** equally likely outcomes, written as ordered pairs (blue, grey).

Why is (1, 4) different from (4, 1)?

(1, 4) means **1 on the blue die and 4 on the grey die**; (4, 1) means 4 on blue and 1 on grey. They are two different ways the dice can land. Throwing one die twice is the same experiment (first throw, second throw).

🎲🎲 Fig. 14.3 as a tool: the 36 outcomes

blue ↓ grey →	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	**(2, 6)**
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	**(3, 5)**	(3, 6)

4	(4, 1)	(4, 2)	(4, 3)	**(4, 4)**	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	**(5, 3)**	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	**(6, 2)**	(6, 3)	(6, 4)	(6, 5)	(6, 6)

E = "sum = 8": 5 favourable outcomes out of 36. P(E) = $\frac{5}{36}$ (≈ 0.139)

(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)

Example 13 (i) · sum 8

(2, 6), (3, 5), (4, 4), (5, 3), (6, 2): 5 outcomes.

P(E) = $\frac{5}{36}$

(ii) · sum 13

The largest possible sum is 6 + 6 = 12. No outcome.

P(F) = $\frac{0}{36}$ = 0 (impossible)

(iii) · sum ≤ 12

Every one of the 36 outcomes.

P(G) = $\frac{36}{36}$ = 1 (sure)

The sum of two dice (Exercise 22)

Sums are **not** equally likely: 7 can be made in 6 ways, but 2 and 12 in only one way each. Read the sums along the diagonals of the grid.

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1
P	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

🧠 Must-know two-dice counts (out of 36)

Event	Count	P
Doublet (same number on both)	6	$\frac{1}{6}$
Different numbers	30	$\frac{5}{6}$
Sum is 7 (the most likely sum)	6	$\frac{1}{6}$
Sum is a prime (2, 3, 5, 7, 11)	15	$\frac{5}{12}$
Sum is even (or odd)	18	$\frac{1}{2}$
Difference is 2	8	$\frac{2}{9}$
5 at least once / 5 neither time	11 / 25	$\frac{11}{36}$ / $\frac{25}{36}$
Product is even	27	$\frac{3}{4}$
Product is 6	4	$\frac{1}{9}$



Probability by length and area (not for the exam)

So far the number of outcomes was **finite**. If the outcome can be **any point** on a line, in a circle or a rectangle, there are infinitely many outcomes and we cannot count them. Instead we compare **lengths** or **areas**:

$$P(E) = \frac{\text{favourable length (or area)}}{\text{total length (or area)}}$$

Examples 10, 11 and Exercise 20 are marked * "Not from the examination point of view".

Example 10 · musical chairs



Fig. 14.1

Fig. 14.1

Music stops at any time in 2 minutes. Stopping in the first half-minute: length $\frac{1}{2}$ out of 2.

$$P(E) = \frac{1}{2} \div 2 = \frac{1}{4}$$

Example 11 · missing helicopter

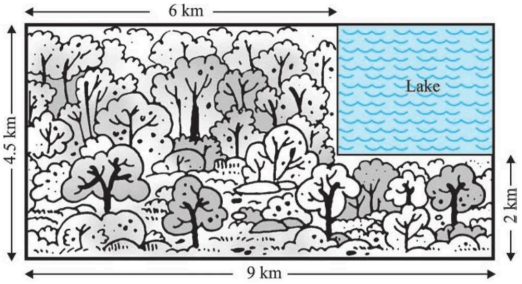


Fig. 14.2

Fig. 14.2 Lake = (9 − 6) × (4.5 − 2) = 3 × 2.5 km^2

Region = 9 × 4.5 = 40.5 km^2. Lake = 3 × 2.5 = 7.5 km^2.

$$P(\text{lake}) = \frac{7.5}{40.5} = \frac{75}{405} = \frac{5}{27}$$

Lab

Experiment vs theory: a note to the reader

📖 A note to the reader (NCERT)

The **experimental** (empirical) probability is based on what **has actually happened**; the **theoretical** probability tries to **predict** what will happen on the basis of certain assumptions. As the **number of trials increases**, we may expect the experimental and theoretical probabilities to be **nearly the same**.

Numbers in a box

Discs or cards numbered 1 to n are another favourite. Know how many primes, squares and multiples there are.

Tables

Tables to learn (1- and 2-mark favourites)

Experimental vs theoretical probability

Point	Experimental (empirical)	Theoretical (classical)
Based on	What **actually happened** in trials	**Assumptions** (equally likely outcomes)
Formula	$\frac{\text{trials in which E happened}}{\text{total trials}}$	$\frac{\text{favourable outcomes}}{\text{all possible outcomes}}$
Needs the experiment?	Yes, repeated many times	No, it is worked out in advance
Value	Can change from one set of trials to another	Fixed for the experiment
Example	60 heads in 100 tosses: $\frac{60}{100} = 0.6$	P(head) = $\frac{1}{2}$ always
Given by	Class IX approach	Laplace, 1795

Kinds of events

Event	Meaning	Probability	Example
Sure (certain)	Always happens	1	A number less than 7 on a die
Impossible	Can never happen	0	Sum 13 with two dice; an orange candy from a bag of lemon candies
Elementary	Has exactly **one** outcome	$\frac{1}{n}$ (n equally likely outcomes)	Getting 4 on a die
Compound (not elementary)	Has two or more outcomes	between 0 and 1	Getting an even number

Complementary (E and $\bar{E}$)	"E" and "not E"; together they cover everything	$P(E) + P(\bar{E}) = 1$	Ace / not an ace
Equally likely	Each has the same chance	equal values	H and T of a fair coin

Sample spaces to remember

Experiment	Total outcomes	Outcomes
One coin	2	H, T
Two coins (or one coin twice)	4	HH, HT, TH, TT
Three coins (or one coin 3 times)	8	HHH, HHT, HTH, THH, HTT, THT, TTH, TTT
One die	6	1, 2, 3, 4, 5, 6
Two dice (or one die twice)	36	(1, 1) ... (6, 6), see Fig. 14.3
A deck of cards	52	4 suits × 13
A letter of the English alphabet	26	5 vowels, 21 consonants
A day of the week / a month of the year	7 / 12	
A non-leap year / a leap year	52 weeks + 1 day / 52 weeks + 2 days	P(53 Mondays) = $\frac{1}{7}$ / $\frac{2}{7}$

Handy counts

Numbers 1 to ...	Primes	Perfect squares	Even
1 to 6 (a die)	3 (2, 3, 5)	2 (1, 4)	3
1 to 20	8 (2, 3, 5, 7, 11, 13, 17, 19)	4	10
1 to 50	15	7	25
1 to 100	25	10	50

1 is **neither prime nor composite**. 2 is the only even prime.

Traps Common mistakes (and Exercise 25)

❌ Counting results that are not equally likely

"Two coins give three outcomes: two heads, two tails, one of each, so each is $\frac{1}{3}$." **Wrong.** "One of each" happens in 2 ways (HT, TH): P = $\frac{2}{4} = \frac{1}{2}$, while two heads and two tails are $\frac{1}{4}$ each. The same mistake: "sums 2 to 12 each have P = $\frac{1}{11}$" (Exercise 22).

✅ Grouping correctly

"A die gives an odd or an even number, so P(odd) = $\frac{1}{2}$." **Correct**, because odd (1, 3, 5) and even (2, 4, 6) each contain 3 of the 6 equally likely outcomes.

❌ "Between" includes the ends

"A number lying between 2 and 6" on a die is 3, 4, 5 only (Exercise 13).

❌ Forgetting to change the total

Without replacement, the total goes down by 1 (Exercises 15 and 17). After a lost card, 51 cards remain.

❌ Treating (1, 4) and (4, 1) as the same

Two dice always give 36 ordered outcomes, not 21.

❌ Leaving the fraction unsimplified

Write $\frac{12}{52} = \frac{3}{13}$. Board examiners expect the lowest form.

CBSE Previous year board questions

Real questions from the CBSE Class 10 Mathematics (Standard) board papers, set 30/1/1, with answers. Filter by year.

– 2026 · 30/1/1 · 1 mark

A die is thrown once. Probability of getting a number other than 3 is: (a) $\frac{1}{6}$ (b) $\frac{3}{6}$ (c) $\frac{5}{6}$ (d) 1

(c) $\frac{5}{6}$. P(3) = $\frac{1}{6}$, so P(not 3) = 1 − $\frac{1}{6} = \frac{5}{6}$. (Or: 1, 2, 4, 5, 6 are 5 favourable outcomes out of 6.)

– 2026 · 30/1/1 · 1 mark

Assertion (A): The probability that a leap year has 53 Mondays is $\frac{2}{7}$. Reason (R): The probability that a non-leap year has 53 Mondays is $\frac{5}{7}$.

(c) A is true, but R is false. A leap year has 366 days = 52 weeks + 2 extra days. The 2 extra days can be (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun): 7 outcomes, 2 contain Monday, so P = $\frac{2}{7}$. A non-leap year has 365 days = 52 weeks + 1 day, which is a Monday in 1 of 7 cases: P = $\frac{1}{7}$, not $\frac{5}{7}$.

– 2026 · 30/1/1 · 3 marks

Two dice of different colours are thrown at the same time. Write down all the possible outcomes. What is the probability that (i) the same number appears on both dice? (ii) different numbers appear on both dice?

The 36 outcomes are (1, 1), (1, 2), ... (6, 6), as in the grid of Fig. 14.3. (i) Same number: (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) = 6. **P** = $\frac{6}{36}$ = $\frac{1}{6}$. (ii) Different numbers is the complement: **P = 1 −** $\frac{1}{6}$ = $\frac{5}{6}$ (30 outcomes).

– 2025 · 30/1/1 · 1 mark

A card is selected at random from a deck of 52 playing cards. The probability of it being a red face card is: (A) $\frac{3}{13}$ **(B)** $\frac{2}{13}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{3}{26}$

(D) $\frac{3}{26}$**.** Red face cards: K, Q, J of hearts and of diamonds = 6. P = $\frac{6}{52}$ = $\frac{3}{26}$.

– 2025 · 30/1/1 · 1 mark

Assertion (A): The probability of selecting a number at random from the numbers 1 to 20 is 1. Reason (R): For any event E, if P(E) = 1, then E is called a sure event.

(A) Both A and R are true and R is the correct explanation of A. Whatever is selected will certainly be one of the numbers 1 to 20, so the event is sure: P = $\frac{20}{20}$ = 1. R is the definition of a sure event.

– **2025 · 30/1/1 · 3 marks**

Two dice are thrown at the same time. Determine the probability that the difference of the numbers on the two dice is 2.

Total outcomes = 36. Difference 2: (1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4) = 8. **P** = $\frac{8}{36} = \frac{2}{9}$. Try it in the two-dice tool (event: difference = 2).

– **2024 · 30/1/1 · 1 mark**

If the probability of a player winning a game is 0.79, then the probability of his losing the same game is: (A) 1.79 (B) 0.31 (C) 0.21% (D) 0.21

(D) 0.21. Winning and losing are complementary: 1 − 0.79 = 0.21. (0.21% would be 0.0021.)

– **2024 · 30/1/1 · 1 mark (reworded)**

From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting a prime number from the remaining is: (A) $\frac{1}{5}$ (B) $\frac{1}{7}$ (C) $\frac{2}{5}$ (D) $\frac{2}{7}$

(A) $\frac{1}{5}$. Remove 4 and 16. Remaining: 1, 7, 9, 21, 25 (5 numbers). Only 7 is prime (1 is not prime). P = $\frac{1}{5}$. (The printed options are reconstructed; the answer is from the solved paper.)

– **2024 · 30/1/1 · 1 mark**

Two dice are rolled together. The probability of getting the sum of the numbers on the two dice as 2, 3 or 5 is: (A) $\frac{7}{36}$ (B) $\frac{11}{36}$ (C) $\frac{5}{36}$ (D) $\frac{4}{9}$

(A) $\frac{7}{36}$. Sum 2: (1, 1) → 1 way. Sum 3: (1, 2), (2, 1) → 2. Sum 5: (1, 4), (4, 1), (2, 3), (3, 2) → 4. Total 7 out of 36.

– **2024 · 30/1/1 · 2 marks**

In a pack of 52 playing cards one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is the queen of hearts, if the lost card is a black card.

Cards left = 52 − 1 = **51**. The lost card is black, so the queen of hearts (a red card) is still in the pack: 1 favourable outcome. **P** = $\frac{1}{51}$.

– 2024 · 30/1/1 · case study, 1 mark

BINGO is played with balls numbered 1 to 75. When the first ball was picked, what was the probability of calling out an even number?

Even numbers from 1 to 75: 2, 4, ..., 74 = **37**. **P** = $\frac{37}{75}$. (The rest of this case study was on the median and mode of the called numbers: Chapter 13.)

– 2023 · 30/1/1 · 1 mark

Probability of happening of an event is denoted by p and probability of non-happening of the event is denoted by q. Relation between p and q is: (A) p + q = 1 (B) p = q = 1 (C) p = q − 1 (D) p + q + 1 = 0

(A) p + q = 1. P(E) + P(not E) = 1.

– 2023 · 30/1/1 · 1 mark

A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought? (A) 40 (B) 240 (C) 480 (D) 750

(C) 480. $\frac{x}{6000} = 0.08 \Rightarrow x = 0.08 \times 6000 = 480$.

– 2023 · 30/1/1 · 1 mark

In a group of 20 people, 5 can't swim. If one person is selected at random, then the probability that he/she can swim is: (A) $\frac{3}{4}$ (B) $\frac{1}{3}$ (C) 1 (D) $\frac{1}{4}$

(A) $\frac{3}{4}$. Swimmers = 20 − 5 = 15. P = $\frac{15}{20} = \frac{3}{4}$.

– 2023 · 30/1/1 · 2 marks

A bag contains 4 red, 3 blue and 2 yellow balls. One ball is drawn at random from the bag. Find the probability that the drawn ball is (i) red (ii) yellow.

Total balls = 4 + 3 + 2 = 9. (i) **P(red)** = $\frac{4}{9}$ (1 mark). (ii) **P(yellow)** = $\frac{2}{9}$ (1 mark). Try it in the bag tool.

Practice Board-pattern practice

MCQs (1 mark)

M1 A letter of the English alphabet is chosen at random. The probability that it is a vowel is (a) $\frac{5}{26}$ (b) $\frac{21}{26}$ (c) $\frac{1}{26}$ (d) $\frac{1}{5}$

(a) $\frac{5}{26}$ (a, e, i, o, u).

M2 Three coins are tossed together. P(exactly two heads) is (a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

(b) $\frac{3}{8}$: HHT, HTH, THH.

M3 If P(E) = 0.3, then P(not E) is (a) 0.3 (b) 0.7 (c) 1.3 (d) −0.3

(b) 0.7

M4 A card is drawn from a deck. P(it is neither a king nor a queen) is (a) $\frac{2}{13}$ (b) $\frac{11}{13}$ (c) $\frac{1}{13}$ (d) $\frac{12}{13}$

(b) $\frac{11}{13}$: $52 - 8 = 44$, $\frac{44}{52} = \frac{11}{13}$.

M5 A die is thrown once. P(a perfect square number) is (a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

(b) $\frac{1}{3}$: 1 and 4.

Assertion–Reason

(a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

AR1 A: When two coins are tossed, P(one head and one tail) = $\frac{1}{2}$. R: Of the 4 equally likely outcomes HH, HT, TH, TT, two have one head and one tail.

(a) $\frac{2}{4} = \frac{1}{2}$. (If someone says $\frac{1}{3}$, they have wrongly taken HT and TH as one outcome.)

AR2 A: The probability of getting 7 on a die is 0. R: The probability of an impossible event is 0.

(a)

– **AR3** **A: P(E) = −0.4 is not possible. R: 0 ≤ P(E) ≤ 1 for every event E.**

(a)

Short answer (2–3 marks)

– **S1** **A box has cards numbered 3, 5, 7, ..., 35, mixed well. One card is drawn. Find P(the number is a prime) and P(a multiple of 7).**

Numbers: 3 to 35 odd: n = (35 − 3)/2 + 1 = **17**. Primes: 3, 5, 7, 11, 13, 17, 19, 23, 29, 31 = 10 → $\frac{10}{17}$.
Multiples of 7: 7, 21, 35 = 3 → $\frac{3}{17}$.

– **S2** **A bag has 5 red and some blue balls. If P(blue) is double P(red), find the number of blue balls.**

Let blue = x. $\frac{x}{5+x} = 2 \times \frac{5}{5+x} \Rightarrow x = 10$. **10 blue balls.**

– **S3** **Two dice are thrown together. Find P(the product of the numbers is a perfect square).**

Products 1, 4, 9, 16, 25, 36: (1, 1), (1, 4), (4, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) = 8. **P** = $\frac{8}{36} = \frac{2}{9}$.

Case study (4 marks)

– **C1** **At the Motihari school fair, a stall has a box of 50 tokens numbered 1 to 50. You win a pen if your token is a multiple of 5, and a chocolate if it is a perfect square. (i) P(winning a pen)? (ii) P(winning a chocolate)? (iii) P(winning both)? OR P(winning nothing)?**

(i) Multiples of 5: 10 → $\frac{1}{5}$. (ii) Squares 1, 4, 9, 16, 25, 36, 49: 7 → $\frac{7}{50}$. (iii) Both: 25 only → $\frac{1}{50}$. OR
Winning something = 10 + 7 − 1 = 16 tokens, so nothing = 34 → $\frac{34}{50} = \frac{17}{25}$.

Recap

Summary (NCERT 14.2)

1. The **theoretical (classical) probability** of an event E is P(E) = $\frac{\text{number of outcomes favourable to E}}{\text{number of all possible outcomes}}$, where the outcomes are equally likely.
2. The probability of a **sure (certain) event is 1**.
3. The probability of an **impossible event is 0**.
4. For every event E, **0 ≤ P(E) ≤ 1**.

5 An event with only one outcome is an **elementary event**. The sum of the probabilities of all the elementary events of an experiment is **1**.

6 For any event E, **P(E) + P($\bar{E}$) = 1**; E and $\bar{E}$ are **complementary events**.

Quiz Test yourself

Tap an answer. **Score: 0 / 20**

1. P(E) + P(not E) =

- 0
- **1**
- ½
- 2

Complementary events (Exercise 1).

2. Which cannot be the probability of an event?

- 2/3
- **−1.5**
- 15%
- 0.7

A probability is never negative (Exercise 4).

3. If P(E) = 0.05, then P(not E) is

- 0.05
- 0.5
- **0.95**
- 1.05

1 − 0.05 (Exercise 5).

4. The probability of an impossible event is

- **0**
- 1
- −1
- not defined

No favourable outcome.

5. A bag has only lemon candies. P(lemon candy) is

- 0
- ½
- **1**
- cannot say

A sure event (Exercise 6).

6. A die is thrown. P(a prime number) is

- 1/3
- **1/2**
- 2/3
- 1/6

2, 3, 5 (Exercise 13).

7. A die is thrown. P(a number between 2 and 6) is

- 2/3
- 5/6
- **1/2**
- 1/3

3, 4, 5 only.

8. One card from a deck. P(a face card) is

- 1/13
- **3/13**
- 4/13
- 3/26

12 face cards out of 52.

9. One card from a deck. P(red face card) is

- 3/13
- 1/26
- **3/26**
- 1/13

6 out of 52 (CBSE 2025).

10. One card from a deck. P(the queen of diamonds) is

- **1/52**
- 1/13
- 1/4
- 1/26

Only one such card.

11. Two coins are tossed. P(at least one head) is

- 1/4
- 1/2
- **3/4**
- 1

HH, HT, TH (Example 9).

12. Two dice are thrown. The number of possible outcomes is

- 12
- 21
- **36**
- 11

6 × 6.

13. Two dice. P(sum is 8) is

- 1/9
- **5/36**
- 1/6
- 2/9

(2,6), (3,5), (4,4), (5,3), (6,2).

14. Two dice. P(5 will not come up either time) is

- 11/36
- **25/36**
- 5/6
- 1/36

5 × 5 = 25 outcomes without a 5 (Exercise 24).

15. A coin is tossed 3 times. P(Hanif loses, i.e. not all same) is

- 1/4
- 1/2
- **3/4**
- 7/8

6 of 8 outcomes (Exercise 23).

16. Discs 1 to 90. P(a perfect square) is

- **1/10**
- 1/9
- 9/10
- 1/5

1, 4, ..., 81: 9 numbers (Exercise 18).

17. 3 red and 5 black balls. P(not red) is

- 3/8
- **5/8**
- 3/5
- 1/2

Exercise 8.

18. Theoretical probability was defined by

- J. Cardan
- James Bernoulli
- **Pierre Simon Laplace**
- de Moivre

In 1795.

19. A leap year is chosen. P(it has 53 Sundays) is

- 1/7
- **2/7**
- 5/7
- 53/366

2 extra days, 7 possible pairs.

20. P(winning a lottery) is 0.08 and 6000 tickets are sold. Tickets bought:

40	240
480	750

0.08 × 6000 (CBSE 2023).

Answers to Exercise 14.1 (all 25 questions)

Every question of the exercise with an answer you can write in the exam. Tap to open.

1. Complete: (i) P(E) + P(not E) = ___ (ii) The probability of an event that cannot happen is ___; such an event is called ___ (iii) The probability of an event that is certain to happen is ___; such an event is called ___ (iv) The sum of the probabilities of all the elementary events of an experiment is ___ (v) The probability of an event is greater than or equal to ___ and less than or equal to ___.

(i) **1** (ii) **0**, an **impossible event** (iii) **1**, a **sure (certain) event** (iv) **1** (v) **0** and **1**.

2. Which of the following experiments have equally likely outcomes? Explain. (i) A driver attempts to start a car: it starts or does not start. (ii) A player attempts to shoot a basketball: she/he shoots or misses. (iii) A trial is made to answer a true-false question: right or wrong. (iv) A baby is born: boy or girl.

(i) **Not equally likely**: whether the car starts depends on factors such as fuel, battery and the condition of the engine; normally it starts. (ii) **Not equally likely**: the result depends on the player's skill and practice. (iii) **Equally likely**: there are two possible answers and (when guessing) each is equally likely to be right. (iv) **Equally likely**: a newborn is equally likely to be a boy or a girl.

3. Why is tossing a coin considered to be a fair way of deciding which team should get the ball at the beginning of a football game?

A fair coin is unbiased, so head and tail are **equally likely**: P(head) = P(tail) = $\frac{1}{2}$. The result cannot be predicted or controlled by either captain, so each team has an **equal chance** of getting the ball.

4. **Which of the following cannot be the probability of an event? (A) $\frac{2}{3}$ (B) −1.5 (C) 15% (D) 0.7**

(B) −1.5, because 0 ≤ P(E) ≤ 1 and a probability can never be negative. (15% = 0.15 is fine.)

5. **If P(E) = 0.05, what is the probability of 'not E'?**

$P(\bar{E}) = 1 - P(E) = 1 - 0.05 =$ **0.95**.

6. **A bag contains lemon flavoured candies only. Malini takes out one candy without looking into the bag. What is the probability that she takes out (i) an orange flavoured candy? (ii) a lemon flavoured candy?**

(i) There is no orange candy: an impossible event, **P = 0**. (ii) Every candy is lemon: a sure event, **P = 1**.

7. **It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992. What is the probability that the 2 students have the same birthday?**

"Same birthday" is the complement of "not the same birthday". P = 1 − 0.992 = **0.008**.

8. **A bag contains 3 red balls and 5 black balls. A ball is drawn at random. What is the probability that the ball drawn is (i) red? (ii) not red?**

Total = 3 + 5 = 8. (i) **P(red)** = $\frac{3}{8}$. (ii) **P(not red)** = $1 - \frac{3}{8} = \frac{5}{8}$.

9. **A box contains 5 red marbles, 8 white marbles and 4 green marbles. One marble is taken out at random. What is the probability that it will be (i) red? (ii) white? (iii) not green?**

Total = 5 + 8 + 4 = 17. (i) $\frac{5}{17}$ (ii) $\frac{8}{17}$ (iii) P(green) = $\frac{4}{17}$, so P(not green) = $1 - \frac{4}{17} = \frac{13}{17}$.

10. **A piggy bank contains hundred 50 p coins, fifty ₹1 coins, twenty ₹2 coins and ten ₹5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a 50 p coin? (ii) will not be a ₹5 coin?**

Total coins = 100 + 50 + 20 + 10 = 180. (i) P(50 p) = $\frac{100}{180} = \frac{5}{9}$. (ii) P(₹5) = $\frac{10}{180} = \frac{1}{18}$, so P(not ₹5) = $1 - \frac{1}{18} = \frac{17}{18}$.

11. **Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish (Fig. 14.4). What is the probability that the fish taken out is a male fish?**

Total = 5 + 8 = 13. **P(male) =** $\frac{5}{13}$.

12. **A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1 to 8 (Fig. 14.5), equally likely. What is the probability that it will point at (i) 8? (ii) an odd number? (iii) a number greater than 2? (iv) a number less than 9?**

Total = 8. (i) $\frac{1}{8}$. (ii) 1, 3, 5, 7: $\frac{4}{8} = \frac{1}{2}$. (iii) 3, 4, 5, 6, 7, 8: $\frac{6}{8} = \frac{3}{4}$. (iv) all 8 numbers: $\frac{8}{8} = 1$ (sure event). Try it on the spinner.

13. **A die is thrown once. Find the probability of getting (i) a prime number (ii) a number lying between 2 and 6 (iii) an odd number.**

Total = 6. (i) 2, 3, 5: $\frac{3}{6} = \frac{1}{2}$. (ii) 3, 4, 5: $\frac{3}{6} = \frac{1}{2}$. (iii) 1, 3, 5: $\frac{3}{6} = \frac{1}{2}$.

14. **One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (i) a king of red colour (ii) a face card (iii) a red face card (iv) the jack of hearts (v) a spade (vi) the queen of diamonds.**

Total = 52. (i) King of hearts and king of diamonds: $\frac{2}{52} = \frac{1}{26}$. (ii) 12 face cards: $\frac{12}{52} = \frac{3}{13}$. (iii) 6 red face cards: $\frac{6}{52} = \frac{3}{26}$. (iv) $\frac{1}{52}$. (v) 13 spades: $\frac{13}{52} = \frac{1}{4}$. (vi) $\frac{1}{52}$. Check each in the deck explorer.

15. **Five cards, the ten, jack, queen, king and ace of diamonds, are well-shuffled face downwards. One card is picked at random. (i) What is the probability that the card is the queen? (ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?**

(i) Total = 5: **P(queen) =** $\frac{1}{5}$. (ii) The queen is put aside, so 4 cards remain (10, J, K, A). (a) **P(ace) =** $\frac{1}{4}$. (b) There is no queen left: **P(queen) = 0**.

16. **12 defective pens are accidentally mixed with 132 good ones. One pen is taken out at random. Find the probability that it is a good one.**

Total = 12 + 132 = 144. **P(good) =** $\frac{132}{144} = \frac{11}{12}$.

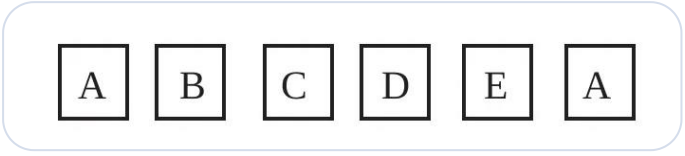
17. **(i) A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random. What is the probability that it is defective? (ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective?**

(i) **P(defective)** = $\frac{4}{20} = \frac{1}{5}$. (ii) A good bulb has been removed: 19 bulbs remain, of which 4 are defective and 16 − 1 = 15 are good. **P(not defective)** = $\frac{15}{19}$.

18. **A box contains 90 discs numbered from 1 to 90. If one disc is drawn at random, find the probability that it bears (i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5.**

Total = 90. (i) Two-digit numbers 10 to 90: 90 − 9 = 81. P = $\frac{81}{90} = \frac{9}{10}$. (ii) Squares 1, 4, 9, 16, 25, 36, 49, 64, 81 = 9. P = $\frac{9}{90} = \frac{1}{10}$. (iii) 5, 10, ..., 90 = 18. P = $\frac{18}{90} = \frac{1}{5}$. See the numbered discs tool.

19. **A child has a die whose six faces show the letters A, B, C, D, E, A. The die is thrown once. What is the probability of getting (i) A? (ii) D?**



Total = 6. (i) A is on 2 faces: $\frac{2}{6} = \frac{1}{3}$. (ii) D is on 1 face: $\frac{1}{6}$.

20*. **Suppose you drop a die at random on the rectangular region shown in Fig. 14.6. What is the probability that it will land inside the circle with diameter 1 m? (Not from the examination point of view.)**

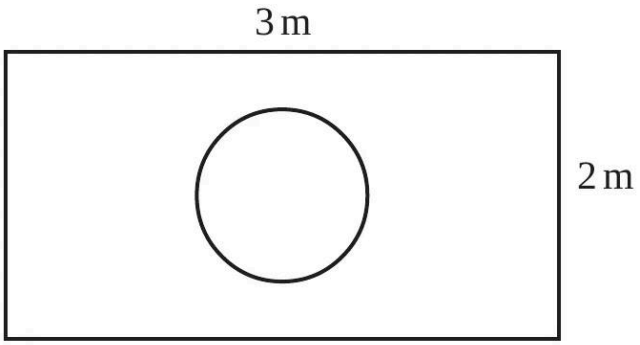


Fig. 14.6

Fig. 14.6

Area of rectangle = $3 \times 2 = 6$ m^2. Radius of circle = 0.5 m, area = $\pi(0.5)^2 = \frac{\pi}{4}$ m^2. **P = $\frac{\pi}{4} \div 6 = \frac{\pi}{24}$** (about 0.13). See the random drop tool.

21. **A lot consists of 144 ball pens of which 20 are defective and the others are good. Nuri will buy a pen if it is good, but not if it is defective. The shopkeeper draws one pen at random. What is the probability that (i) she will buy it? (ii) she will not buy it?**

Good pens = 144 − 20 = 124. (i) P(buy) = $\frac{124}{144} = \frac{31}{36}$. (ii) P(not buy) = $\frac{20}{144} = \frac{5}{36}$ (or $1 - \frac{31}{36}$).

22. **Refer to Example 13. (i) Complete the table of the probabilities of the sums 2 to 12. (ii) A student argues that there are 11 possible outcomes 2, 3, ..., 12, so each has probability $\frac{1}{11}$. Do you agree? Justify.**

(i)

Sum	2	3	4	5	6	7	8	9	10	11	12
P	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(ii) **No.** The 11 sums are **not equally likely**: a sum of 7 can occur in 6 ways but a sum of 2 in only 1 way. The equally likely outcomes are the 36 ordered pairs, so P(sum 2) = $\frac{1}{36}$, P(sum 7) = $\frac{6}{36}$, and so on.

23. **A game consists of tossing a one rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result (three heads or three tails) and loses otherwise. Calculate the probability that Hanif will lose the game.**

Outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT = 8. Win: HHH, TTT = 2, so P(win) = $\frac{2}{8} = \frac{1}{4}$. **P(lose) = 1 − $\frac{1}{4}$ = $\frac{3}{4}$** (6 outcomes). Use the coin tree with 3 coins.

24. **A die is thrown twice. What is the probability that (i) 5 will not come up either time? (ii) 5 will come up at least once?**

Throwing a die twice is the same as throwing two dice: 36 outcomes. (i) Outcomes with no 5: each throw has 5 choices (1, 2, 3, 4, 6), so 5 × 5 = 25. **P = $\frac{25}{36}$**. (ii) Complement of (i): **P = 1 − $\frac{25}{36}$ = $\frac{11}{36}$**. (The 11 outcomes: (5, 1) ... (5, 6) and (1, 5), (2, 5), (3, 5), (4, 5), (6, 5).)

25. **Which arguments are correct and which are not? Give reasons. (i) If two coins are tossed, there are three possible outcomes: two heads, two tails or one of each, so each has probability $\frac{1}{3}$. (ii) If a die is thrown, there are two possible outcomes, odd or even, so P(odd) = $\frac{1}{2}$.**

(i) **Incorrect.** The equally likely outcomes are HH, HT, TH, TT. "One of each" happens in 2 ways, so P(two heads) = $\frac{1}{4}$, P(two tails) = $\frac{1}{4}$, P(one of each) = $\frac{2}{4} = \frac{1}{2}$. (ii) **Correct.** Odd numbers 1, 3, 5 and even numbers 2, 4, 6 are 3 each out of 6 equally likely outcomes, so P(odd) = $\frac{3}{6} = \frac{1}{2}$.

The "Why?" questions inside the chapter

i **Example 1: why is P(tail) = $\frac{1}{2}$?**

Tail is 1 of the 2 equally likely outcomes (H, T). Also P(tail) = 1 − P(head) = 1 − $\frac{1}{2}$.

ii **Example 4: why are there 52 possible outcomes, and why are 48 favourable to "not an ace"?**

Each of the 52 cards is equally likely to be drawn after shuffling, so there are 52 outcomes. 4 of them are aces, so 52 − 4 = 48 cards are not aces.

iii **Example 7: why are 25 outcomes favourable for a girl and 15 for a boy?**

There is one identical card for each student: 25 cards have girls' names and 15 have boys' names.

iv Example 8: why is the number of possible outcomes 3 + 2 + 4 = 9?

Each of the 9 marbles is equally likely to be drawn (that is what "at random" means), so each marble is one outcome, whatever its colour.

v Example 9: why are (H, H), (H, T), (T, H) favourable to "at least one head"?

"At least one head" means one head or two heads. Only (T, T) has no head; the other three outcomes have at least one head.

vi Example 12: why are 96 shirts acceptable to Sujatha?

She rejects only the 4 shirts with major defects, so she accepts the 88 good and the 8 with minor defects: 88 + 8 = 96 (= 100 − 4).

vii Example 13: why is the pair (1, 4) different from (4, 1)?

The first number is on the blue die and the second on the grey die. (1, 4) is 1 on blue and 4 on grey; (4, 1) is 4 on blue and 1 on grey: two different results.

Figures from the NCERT Mathematics textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Mathematics (Standard) papers 2023 to 2026.